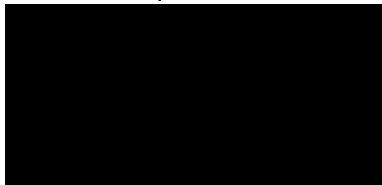


University of Waterloo
Faculty of Engineering
Department of Electrical and Computer Engineering

Behaviour-Driven Drug Adherence Application

Group 2017.025



Consultant:
Derek Rayside

May 28, 2016

Table of Contents

1. High Level Description	3
1.1. Motivation	3
1.2. Project Objective	3
1.3. Block Diagram	4
1.3.1 Mobile Client	5
1.3.2 Backend Services	5
2. Project Specifications	5
2.1 Functional Specifications	6
2.2 Non-functional Specifications	7
3. Risk Assessment	8
4. References	9

1. High Level Description

1.1. Motivation

Failure to take medication is an expensive mistake many patients make, costing the United States an estimated \$100 billion a year [1] . Beyond this, drug efficacy is drastically reduced when medicine is not taken with proper schedules and instruction. HIV-positive patients' viral loads are capable of infecting others, even if only a single pill is missed or forgotten, for example [2]. This, however, is overlooked by most patients as an important part of care. The proposed application called Medicoo will attempt to increase drug adherence by following user patterns and previous studies into drug adherence [3]. Medication is often taken very regularly, and any disruption in a patient's daily pattern can affect their ability to take their medication on time. Therefore there is a need to push more frequent reminders to patients when they are changing their schedule in order to ensure that they are taking their medication on time.

1.2. Project Objective

The goal of Medicoo is to create a system of reminders based on users' behaviour. At a minimum, the user can set a medication schedule that will remind them to take their medication. People often forget medication because something unforeseen occurs. Medicoo will attempt to use a user's data to predict when they will deviate from their regular behaviour and remind them to take their medication. This may be based on a variety of factors like location, time or schedule. In this way, even abnormal schedules for a user will not prevent them from adhering to their medication schedule, hopefully increasing their drug adherence and health.

Medicoo has several competitors. There are habit building applications or games that encourage a user to build a specific habit like drinking water or doing homework. One problem with these habit building applications is that they are not specific and do not use a user's data to make suggestions for participating in the habit. Another issue is that it mentally frames the habit as something arduous or difficult, offering a user or player rewards on their completion. In contrast, Medicoo should instill a positive attitude towards medication. Other competitors are specifically for medication. These also only have time based notifications that are not helpful if a user's plans change and, for instance, do not have

access to their medication. Medicoo will attempt to predict things like travel in order to remind users that they should bring their medication.

1.3. Block Diagram

Figure 1 shown below outlines the block diagram of the Medicoo system.

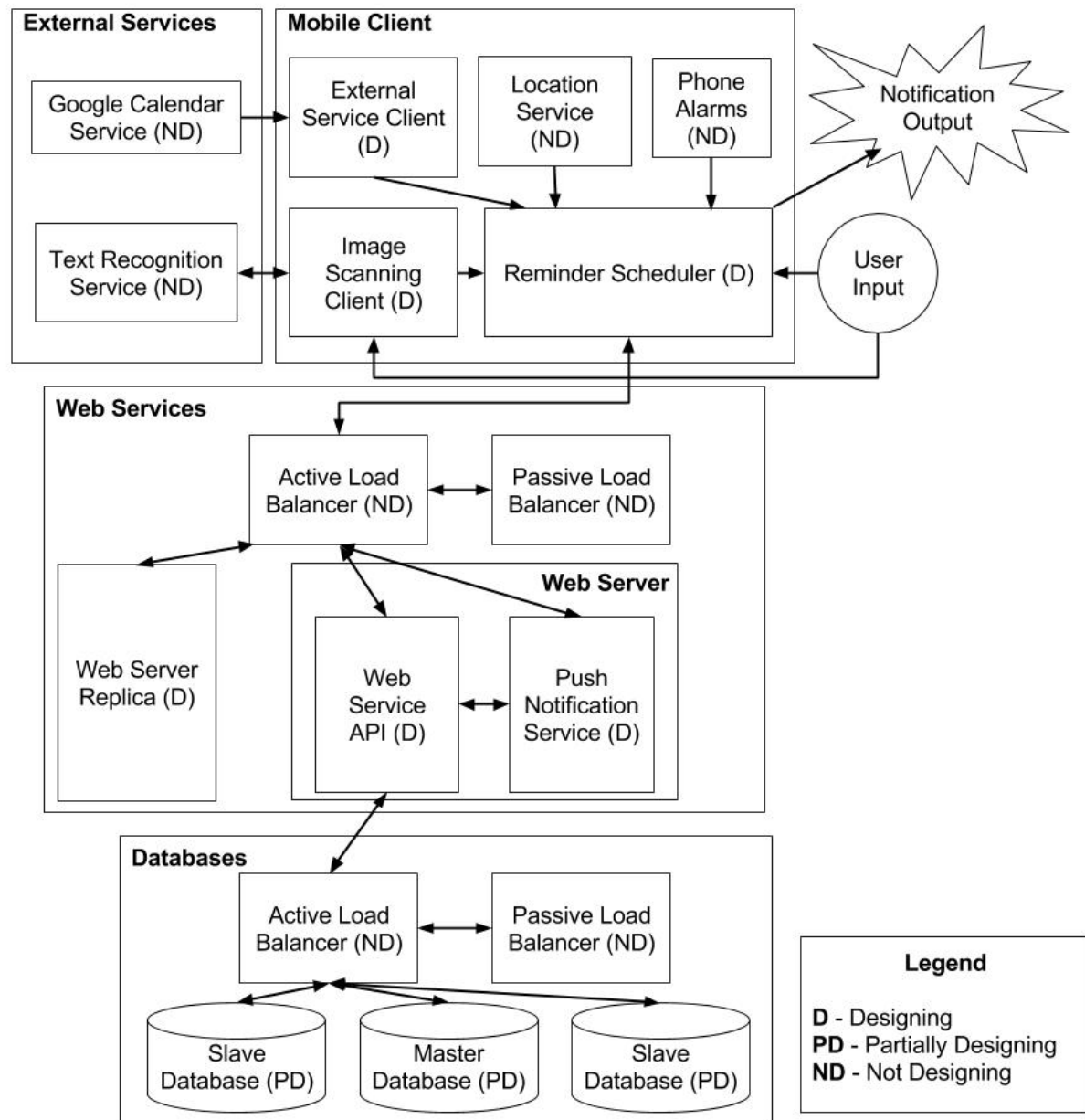


Figure 1: Block diagram of the Medicoo system

1.3.1 Mobile Client

The mobile client will be the centerpiece of the application which provides reminders in the form of notifications. The notifications will be pushed when the user is scheduled to take their medication, or when the user deviates from their normal pattern. The algorithm which will detect these abnormalities will be performed on the mobile device, since the algorithm will required data from various sources. These sources will include the phone's native alarm system, and Google's location and calendar services. The client will also require the user to input the medication schedule to have regular reminders. An additional non-essential feature will be to automatically generate the schedule using image recognition on the medication labels.

1.3.2 Backend Services

The backend service's primary purpose is to provide the user with notifications when the app has not yet been opened. Since the application cannot send notifications from the phone if the app is not enabled, another source must provide timely notifications to the user. In this case, a web server will be user to send push notifications to the mobile client. These notifications will only be based on the user-input medication schedule, since the backend server will not have access to the other external sources. In order to send the notifications at the correct time, the schedules will need to be stored in a database. A slave-master replication pattern will be used on the databases to ensure availability of data and to ensure that no data is lost in the event of disk failure. In order to support multiple users in the future, these backend services could be scaled horizontally, using load balancers in combination with multiple web servers and databases.

2. Project Specifications

The functional and non-functional specifications for Medicoo are presented in the tables below. Table 1 shows the functional specifications, and Table 2 outlines the non-functional specifications. The specifications are classified as essential or non-essential, depending on whether the app will be fully operation with those specifications.

2.1 Functional Specifications

Table 1: Essential and non-essential functional specifications

Specification	Description	Necessity
Supports variable number medication	Medicoo should be able to support a minimum of 10 unique medications and create notifications accordingly.	Essential
Detects and notifies user on behaviour and medication schedule	Medicoo should present the user with a notification when abnormal behaviour is detected, or when user is required to take their medication with 99% accuracy.	Essential
Notification triggers in a timely manner	The notification should happen within 1 minute of medical intake period, or after abnormal behaviour constraints are met.	Essential
Identifies abnormal behavior based on location	Medicoo will detect when the user is more than 20km away from their most traveled locations	Essential
Identifies abnormal behavior based on the user's sleeping patterns	Medicoo will detect when the user wakes or sleeps 2 or more hours from their norm.	Essential
Identifies abnormal behavior based on the user's scheduled events	Medicoo will detect when the user schedules an event that is greater than 8 hours in length.	Essential
Dedicated servers to send user notifications	Medicoo will have dedicated server that can handle 500 requests a minute for the purpose of notifications.	Non-essential
Text recognition in images for medicine labels	Medicoo should have 90% accuracy of scanning drug labels to automatically record data for refills, dosages, and intake schedule	Non-essential
Training behaviour	Medicoo should take no more than two weeks to train on user behaviour and set up notifications	Non-essential

2.2 Non-functional Specifications

Table 2: Essential and non-essential non-functional specifications

Specification	Description	Necessity
Application Size	Medicoo must not exceed 50 MB in size.	Essential
Backend Server Uptime	The backend services must have an uptime of 99%	Essential
Design Standards	Medicoo must follow Google's Material Design pattern	Essential
Data Security	The data store on the databases must be sent and stored securely to ensure privacy of user data.	Essential
Offline usability	The time and location reminders must work without internet access.	Essential
Supportability	Medicoo must be supported on Android 4.2.2 Jelly Bean and higher, giving access to approximately 85% of all android users.	Essential
Scalability	Must be able to support over 100,000 active users at once	Non-essential
Battery Consumption	Medicoo must use at most 5% battery per full charge	Non-essential

3. Risk Assessment

Table 3: Project Risks

Risk	Problem	Solution	Assessment
Technical Knowledge	This project will be heavily reliant on Android programming. Not only that, it will include non trivial implementations and uses of Android services that few in our group have any experience with. This may introduce heavy delays in the implementation of the project and thus reduce scope.	Several things will be done to minimize this impact. Technical documentation will be created and shared between group members in order to minimize onboarding. Difficult implementations will be initially driven by the person with the most experience as.	Impact: Medium Probability: High
Technical Difficulty	Implementing an algorithm to track behaviour and patterns in users is difficult and imperfect. There is a risk that the pattern tracking is inaccurate or unusable. This would make our app much less useful than intended.	To minimize this, the algorithm will be implemented very simply and very early. It will then be iterated upon. If necessary, further research in journals that have used machine learning to study user behaviour will be leveraged.	Impact: High Probability: Medium
Location	Our team will be separated by coop from September to December, 2016. This will mean collaboration will have to occur remotely, reducing productivity.	Strict code reviews and version control will be used for all code that goes into the system.	Impact: Low Probability: Medium
Insufficient Time	Training and testing user behaviour is often a long term process. This extends any form of testing the project will have. Further, many of the features will require separate integrations and must be planned for accordingly. Poor time management may lead to an unfinished project.	Good time management and delegation will be used to ensure project completion. Further, ticket management software like JIRA or Taiga will be leveraged in order to increase transparency about project completion.	Impact: High Probability: Low

References

- [1] L. Osterberg and T. Blaschke, "Drug Therapy: Adherence to Medication," *The New England Journal of Medicine*, vol. 353, no. 5, pp. 487–497, Aug. 2005.
- [2] D. R. Bangsberg et al., "Adherence to protease inhibitors, HIV-1 viral load, and deve...: AIDS," *AIDS*, vol. 14, no. 4, pp. 357–366, Mar. 2000.
- [3] M. Viswanathan et al., "Interventions to improve adherence to self-administered medications for chronic diseases in the United States: A systematic review," *Annals of Internal Medicine*, vol. 157, no. 11, pp. 785–795, Dec. 2012.